LECTURE 6

Review

<u>Divergence Test</u>

If $\lim_{n \to \infty} a_n \neq 0$ then $\sum_{n=1}^{\infty} a_n$ diverges.

<u>Geometric Series Test</u>

For $\sum_{n=1}^{\infty} ar^n$, $\begin{cases} \text{converges if } |r| < 1 \\ \text{diverges if } |r| \geq 1 \end{cases}$

<u>P - series Test</u>

For $\sum_{n=1}^{\infty} \frac{1}{n^p}$, $\begin{cases} \text{converges if } p > 1 \\ \text{diverges if } p \leq 1 \end{cases}$

<u>Integral Test</u>

f is positive, continuous, and decreasing, then

$\lim_{t \to \infty} \int_1^t f(x)\,dx$ converges iff $\sum_{n=1}^{\infty} f(n)$ converges.

Theorem

<u>Limit Comparison Test</u>

If $a_n > 0$, $b_n > 0$, then $\lim_{n \to \infty} \frac{a_n}{b_n} = L$.

(1) If $0 < L < \infty$, then $\sum_{n=1}^{\infty} a_n$ converges iff $\sum_{n=1}^{\infty} b_n$ converges.

(2) If $L = 0$ and $\sum_{n=1}^{\infty} b_n$ converges, then $\sum_{n=1}^{\infty} a_n$ too.

(3) If $\lim_{n \to \infty} a_n = \infty$ and $\sum_{n=1}^{\infty} b_n$ diverges, then $\sum_{n=1}^{\infty} a_n$ too.

limit comparison test

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Ex

Determine if the following converge or diverge.

$\sum_{n=1}^{\infty} \frac{n^2}{n^4+n}$ As $n \to \infty$, we can rewrite as $\frac{1}{n^2}$.

Take $a_n = \frac{n^2}{n^4+n}$, $b_n = \frac{1}{n^2}$.

We have
$$\begin{aligned} L &= \lim_{n\to\infty} \frac{a_n}{b_n} \\ &= \lim_{n\to\infty} \frac{n^2}{n^4+n} \cdot \frac{n^2}{1} \\ &= \lim_{n\to\infty} \frac{n^4}{n^4+n} \\ &= \lim_{n\to\infty} \frac{1}{1+\frac{1}{n^3}} \\ &= 1 \end{aligned}$$

Thus, $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \frac{1}{n^2}$ converges by the p-series test, since $p = 2 > 1$. Hence $\sum_{n=1}^{\infty} \frac{n^2}{n^4+n}$ also converges by the LCT.

$\sum_{n=1}^{\infty} \frac{1}{e^n + \sin(n)}$ Ressembles $\sum_{n=1}^{\infty} \frac{1}{e^n}$.

We have $a_n = \frac{1}{e^n + \sin(n)}$, $b_n = \frac{1}{e^n}$.

Then
$$\begin{aligned} L &= \lim_{n\to\infty} \frac{a_n}{b_n} \\ &= \lim_{n\to\infty} \frac{1}{e^n+\sin(n)} \cdot \frac{e^n}{1} \\ &= \lim_{n\to\infty} \frac{1}{1+\frac{\sin(n)}{e^n}} \\ &= 1 \end{aligned}$$

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Hence $0 < L < \infty$ and $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{1}{e}\right)^n$ converges by the GST since $|r| = \frac{1}{e} < 1$.

Thus, $\sum_{n=1}^{\infty} \frac{1}{e^n + \sin(n)}$ converges by the LCT.

$\sum_{n=2}^{\infty} \frac{1}{\ln(n)}$ We start by trying $b_n = \frac{1}{n}$.

We have $a_n = \frac{1}{\ln(n)}$, $b_n = \frac{1}{n}$. Then,

$$
\begin{aligned}
L &= \lim_{n \to \infty} \frac{a_n}{b_n} \\
&= \lim_{n \to \infty} \frac{1}{\ln(n)} \cdot \frac{n}{1} \\
&\overset{H}{=} \lim_{n \to \infty} \frac{1}{\frac{1}{n}} \\
&= \infty
\end{aligned}
$$

Since $\lim_{n \to \infty} \frac{a_n}{b_n} = \infty$ and the harmonic series diverges, $\sum_{n=1}^{\infty} \frac{1}{\ln(n)}$ diverges by LCT.